

Initial power on and configuration

Starting and configuring the system

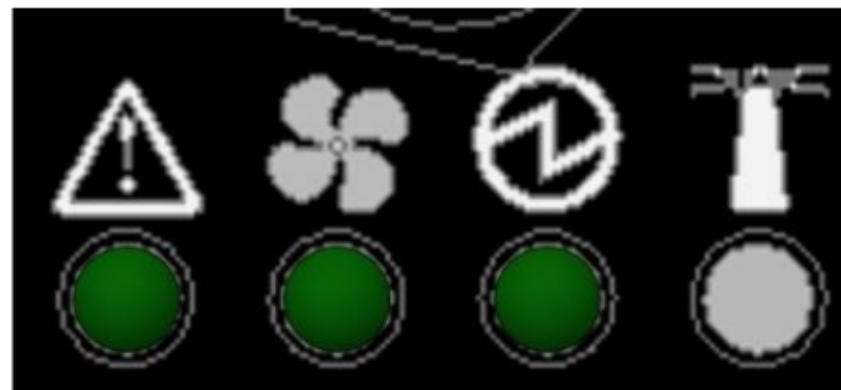
Lenovo

Initial power on

The system platform will automatically power on when AC power is applied. Use standard, three-wire AC power cables that include a safety.

1. Plug in the first power cable.
2. Plug in the second power cable.
3. Wait for the System Status LED to turn green.
4. Check the System Status LEDs and confirm that all the LEDs show status lights consistent with normal operation: initially flashing and then moving to a steady light, as shown in the example picture.

Refer to the Status and port LEDs page in the [Problem determination and troubleshooting](#) section of this course for more information.



Note: It may take up to five minutes for the system to turn on. If the System Status LED is red or amber after five minutes, unplug the system and call a service representative for assistance.

Cable connections to the management port

Typically, switches are shipped from the manufacturer with an RS232 serial cable in the box. As most PCs and laptops do not have an RS232 port, servicers should prepare another USB to serial adapter and install the corresponding driver to manually connect to the switch.



RS232 serial console cable



USB to serial adapter

Connecting to a switch using a serial port

Switches that support Cumulus Linux always have at least one dedicated Ethernet management port, which is named eth0. This interface is geared specifically for out-of-band management use. By default, the management interface uses DHCPv4 for addressing. You can set a static IP address with NCLU or by editing the `/etc/network/interfaces` Linux file. Make sure to connect to the Ethernet management () port.

When starting Cumulus Linux for the first time, the management port will make a DHCPv4 request. To determine the IP address of the switch, you can cross-reference the MAC address of the switch with your DHCP server. The MAC address is typically located on the side of the switch or on the box in which the unit was shipped.

Settings to connect to a switch using a serial port

When connecting to a switch serial port using an RS232 console cable, open a serial terminal program on your host PC using the settings shown in the table below. Make sure the cable is connected to the console RJ45 (10101) port.

Parameter	Settings
Baud rate	115200
Data bits	8
Stop bits	1
Parity	None
Flow control	None

Login credentials

The default installation includes the “root” system account with full system privileges and also the “cumulus” user account with sudo privileges. The root account password is locked by default, which prohibits users from logging in. The default login account name and password for the cumulus account is cumulus/cumulus.

When you log in to Cumulus Linux for the first time with the cumulus account, you will be prompted to change the default password. After you provide a new password, the SSH session will disconnect, and you will have to reconnect with the new password.

Account	root	cumulus
Privilege	Native administrative rights	Administrative rights through the <code>sudo</code> command
Access	Direct login for protocols such as SSH, telnet, and FTP are disabled	SSH and console access allowed
Password	Password disabled	Default Password: <code>cumulus</code>

Logging in for the first time

When Cumulus Linux is started for the first time, the management port will make a DHCP v4 request. To determine the IP address of the switch, you can cross-reference the MAC address of the switch with your DHCP server. The MAC address is typically located on the side of the switch or on the box in which the unit was shipped. You can also access the serial console to configure or check the management port configuration.

- NVUE management interface configuration commands:

```
cumulus@switch:~$ nv set interface eth0 ip address 192.0.2.42/24
cumulus@switch:~$ nv set interface eth0 ip gateway 192.0.2.1
cumulus@switch:~$ nv config apply
```

- Or, edit the `/etc/network/interfaces` file:

```
cumulus@switch:~$ sudo nano /etc/network/interfaces
# Management interface
auto eth0
iface eth0
    address 192.0.2.42/24
    gateway 192.0.2.1
```

Note: Remember to use the `nv config apply` command to set the pending configuration as the applied configuration.

Configuring the hostname

The hostname identifies the switch, so it should be both unique and descriptive.

```
cumulus@switch:~$ nv set system hostname leaf01  
cumulus@switch:~$ nv config apply
```

Note: Underscores (_) are not permitted in the hostname. Apostrophes or non-ASCII characters should also be avoided, as Cumulus Linux does not parse these characters.

Note: The command prompt in the terminal will not reflect the new hostname until you either log out of the switch or start a new shell.

Configuring the time zone

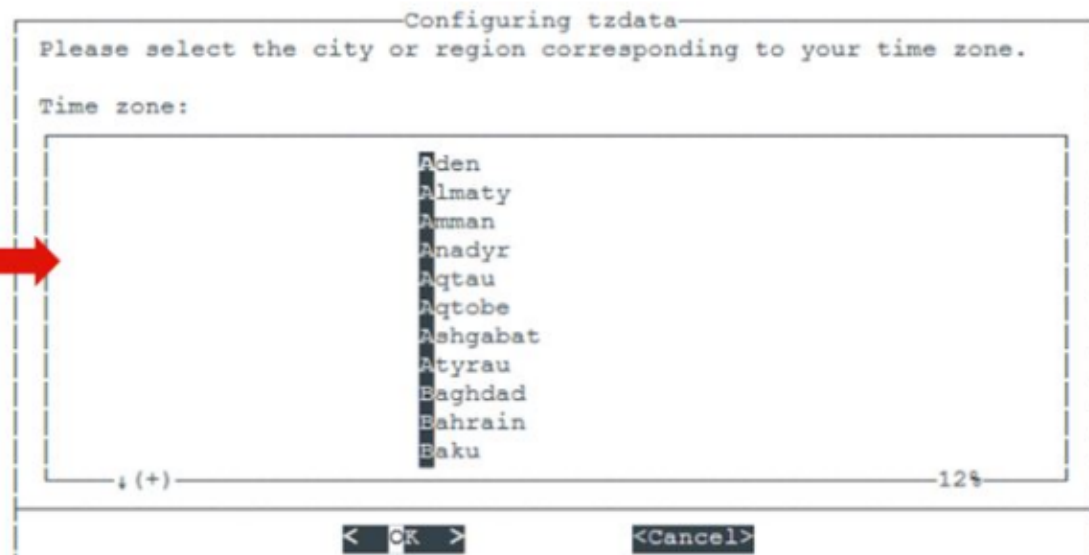
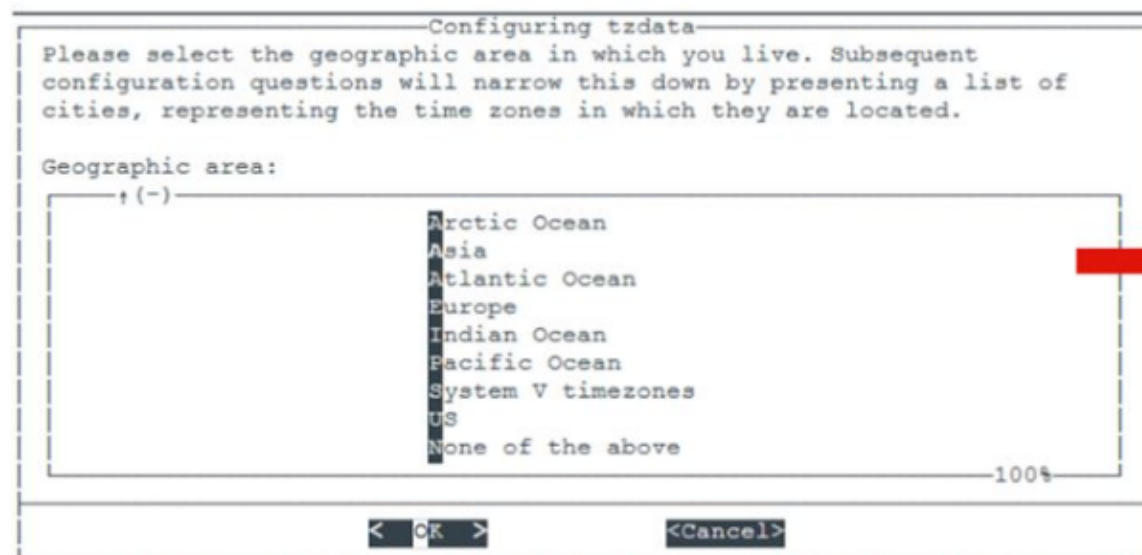
The default time zone on the switch is UTC (Coordinated Universal Time). Change the time zone on your switch to match the time zone in your location.

Example:

```
cumulus@switch:~$ nv set system timezone US/Eastern  
cumulus@switch:~$ nv config apply
```

You can also run the following command in a terminal and follow the onscreen menu options to select the geographic area and region:

```
cumulus@switch:~$ sudo dpkg-reconfigure tzdata
```



Configuring the date and time

Use one of the following methods to configure the date and time:

- Use the following command to set the system clock according to the configured time zone:

```
cumulus@switch:~$ sudo date -s "Tue Jan 26 00:37:13 2023"
```

- Write the current value of the system (software) clock to the hardware clock using the `hwclock` command:

```
cumulus@switch:~$ sudo hwclock -w
```

- Use NTP

The `ntpd` daemon running on the switch implements the NTP protocol. It synchronizes the system time with time servers listed in the `/etc/ntp.conf` file. By default, the `ntpd` daemon is started at boot.